0.1 Spacetime symetries

$\boxed{1.}$ For an infinitesimal boost in the $[0i]$ plane, the spacetime coordinates transform as:

$$t \to t - \eta x^i \qquad \text{and} \qquad x^i \to x^i - \eta t,$$

therefore we get the Killing vector field corresponding to the boost:

$$k^\mu = \frac{\partial x^\mu}{\partial \eta} = (-x^i, -t\delta^\mu_i)^{\mathrm{T}} \implies k = -x^i\partial_t - t\partial_{x^i}$$

Overall we obtain for the conserved current:

$$\begin{aligned}
J_\mu^{[0i]} &= \frac{\partial \mathcal{L}}{\partial(\partial^\mu \phi)} k^\nu \delta_\nu \phi + \frac{\partial \mathcal{L}}{\partial(\partial^\mu \phi^*)} k^\nu \delta_\nu \phi^* - k_\mu \mathcal{L} \\
&= \partial_\mu \phi^* (-x^i\partial_t - t\partial_{x^i})\phi + \partial_\mu \phi (-x^i\partial_t - t\partial_{x^i})\phi^* - k_\mu \mathcal{L} \\
&= -\partial_\mu \phi^* (x^i\partial_t + t\partial_{x^i})\phi - \partial_\mu \phi (x^i\partial_t + t\partial_{x^i})\phi^* - ({\eta_\mu}^i t - {\eta_\mu}^0 x^i)\mathcal{L}
\end{aligned}$$

$\boxed{2.}$ Using the Klein-Gordon equation for each independent field

$$(\partial^\mu\partial_\mu + m^2)\phi = 0 \qquad \text{and} \qquad (\partial^\mu\partial_\mu + m^2)\phi^* = 0$$

we get

$$\begin{aligned}
\partial^\mu J_\mu^{[0i]} = &- (\partial^\mu\partial_\mu\phi^*)(x^i\partial_t + t\partial_{x^i})\phi - \partial_\mu\phi^*(\delta^{\mu i}\partial_t + \delta^{\mu t}\partial_{x^i})\phi - \partial_\mu\phi^*(x^i\partial_t \\
&+ t\partial_{x^i})\partial^\mu\phi - c.c. - ({\eta_\mu}^i\delta^{\mu t} - {\eta_\mu}^0\delta^{\mu i})\mathcal{L} \\
&- ({\eta_\mu}^i t - {\eta_\mu}^0 x^i)\left[\partial_\nu\phi^*\partial^\mu\partial^\nu\phi + m^2\phi^*\partial^\mu\phi\right] - c.c. \\
= &\, 0,
\end{aligned}$$

where the following terms cancel:

- $({\eta_\mu}^i\delta^{\mu t} - {\eta_\mu}^0\delta^{\mu i}) = 0$ using the diagonality of the Minkowski metric
- $-(\partial^\mu\partial_\mu\phi^*)(x^i\partial_t + t\partial_{x^i})\phi - ({\eta_\mu}^i t - {\eta_\mu}^0 x^i)m^2\phi^*\partial^\mu\phi = 0$ using the EOM
- $-\partial_\mu\phi^*(x^i\partial_t + t\partial_{x^i})\partial^\mu\phi - ({\eta_\mu}^i t - {\eta_\mu}^0 x^i)\partial_\nu\phi^*\partial^\mu\partial^\nu\phi = 0$
- the complex conjugate terms similar to the two above cancel each other as well
- $-\partial_\mu\phi^*(\delta^{\mu i}\partial_t + \delta^{\mu t}\partial_{x^i})\phi - \partial_\mu\phi(\delta^{\mu i}\partial_t + \delta^{\mu t}\partial_{x^i})\phi^*$ cancel as we can write $\delta^{\mu i} = -\eta^{\mu i}$ and $\delta^{\mu t} = \eta^{\mu t}$ and the cross terms cancel each other

$\boxed{3.}$ We can write

$$\begin{aligned}\partial^\mu J_\mu^{(A)} &= (\partial^\mu T_{\mu\nu})k^\nu_{(A)} + T_{\mu\nu}(\partial^\mu k^\nu_{(A)}) \\ &= 0 + \frac{1}{2}T_{\mu\nu}\left[\partial^\mu k^\nu_{(A)} + \partial^\nu k^\mu_{(A)}\right] + \frac{1}{2}\left[\partial^\mu k^\nu_{(A)} - \partial^\nu k^\mu_{(A)}\right] \\ &= \frac{1}{2}T_{\mu\nu}\cdot 0 - 0 = 0,\end{aligned}$$

where we have used the conservation of the energy-stress tensor in the first step, the Killing equation and the symmetry of energy-stress tensor in $(\mu\nu)$ in the second step.

In the (optional) part we have the Killing field for translations in direction (ν):

$$x^\mu \to x^\mu + \eta\delta^\mu_{(\nu)} \implies k^\mu_{(\nu)} = \delta^\mu_{(\nu)} \implies k = \delta^\mu_{(\nu)}\partial_\mu$$

and therefore we get:

$$J_{\mu(\nu)} = \partial_\mu\phi^*\delta^\kappa_{(\nu)}\partial_\kappa\phi + \partial_\mu\phi\delta^\kappa_{(\nu)}\partial_\kappa\phi^* - \delta_{\mu(\nu)}\mathcal{L} =: T_{\mu(\nu)}$$

Using this relation we obtain

$$\begin{aligned}J_\mu^{[0i]} &= T_{\mu\nu}k^\nu_{[0i]} = \left[\partial_\mu\phi^*\delta^\kappa_\nu\partial_\kappa\phi + \partial_\mu\phi\delta^\kappa_\nu\partial_\kappa\phi^* - \delta_{\mu\nu}\mathcal{L}\right]\left[-\delta^\nu_t x^i - \delta^\nu_i t\right] \\ &= -\partial_\mu\phi^*(x^i\partial_t + t\partial_{x^i})\phi - \partial_\mu\phi(x^i\partial_t + t\partial_{x^i})\phi^* + (\delta^i_\mu t + \delta^t_\mu x^i)\mathcal{L},\end{aligned}$$

which is the same expression as the one obtained in the first exercise.

$\boxed{4.}$ For the conserved charge we have

$$\begin{aligned}Q^{[0i]} &= \int_{t=\text{const.}} \mathrm{d}^3\boldsymbol{x}\, J_0^{[0i]} \\ &= \int_{t=\text{const.}} \mathrm{d}^3\boldsymbol{x}\, \left[-\partial_t\phi^*(x^i\partial_t + t\partial_{x^i})\phi - \partial_t\phi(x^i\partial_t + t\partial_{x^i})\phi^* - ({\eta_t}^i t - {\eta_t}^t x^i)\mathcal{L}\right] \\ &= -\int_{t=\text{const.}} \mathrm{d}^3\boldsymbol{x}\, x^i\left[\partial_t\phi^*\partial_t\phi + \partial_t\phi\partial_t\phi^* - \mathcal{L}\right] - t\int_{t=\text{const.}} \mathrm{d}^3\boldsymbol{x}\, \left[\partial_t\phi^*\partial_i\phi + \partial_t\phi\partial_i\phi^*\right] \\ &= -\int_{t=\text{const.}} \mathrm{d}^3\boldsymbol{x}\, x^i\mathcal{H} - t\int_{t=\text{const.}} \mathrm{d}^3\boldsymbol{x}\, T_{0i}\end{aligned}$$

$\boxed{5.}$ We can express the fields ϕ and ϕ^* using the creation and annihilation operators for (anti)particles:

$$\phi(x) = \int \tilde{\mathrm{d}}\boldsymbol{k} \left[e^{-ikx}\hat{a}_+(k) + e^{ikx}\hat{a}^\dagger_-(k)\right]$$
$$\phi^*(x) \equiv \phi^\dagger(x) = \int \tilde{\mathrm{d}}\boldsymbol{k} \left[e^{-ikx}\hat{a}_-(k) + e^{ikx}\hat{a}^\dagger_+(k)\right]$$

and therefore also their time derivatives:

$$\pi^\dagger(x) = \partial_t\phi(x) = -i\int \tilde{\mathrm{d}}\boldsymbol{k}\, E(\boldsymbol{k}) \left[e^{-ikx}\hat{a}_+(k) - e^{ikx}\hat{a}^\dagger_-(k)\right]$$
$$\pi(x) = \partial_t\phi^\dagger(x) = -i\int \tilde{\mathrm{d}}\boldsymbol{k}\, E(\boldsymbol{k}) \left[e^{-ikx}\hat{a}_-(k) - e^{ikx}\hat{a}^\dagger_+(k)\right]$$

or in general:

$$\partial_\mu\phi(x) = -i\int \tilde{\mathrm{d}}\boldsymbol{k}\, k_\mu \left[e^{-ikx}\hat{a}_+(k) - e^{ikx}\hat{a}^\dagger_-(k)\right]$$
$$\partial_\mu\phi^\dagger(x) = -i\int \tilde{\mathrm{d}}\boldsymbol{k}\, k_\mu \left[e^{-ikx}\hat{a}_-(k) - e^{ikx}\hat{a}^\dagger_+(k)\right]$$

Since the conserved charge is constant in time t, we can compute it at $t = 0$, therefore only the first term survives. Then we have

$$\begin{aligned}
Q^{[0i]} &= -\int \mathrm{d}^3\boldsymbol{x}\, x^i\mathcal{H} = -\int \mathrm{d}^3\boldsymbol{x}\, x^i \left[\partial_\mu\phi^*\partial^\mu\phi + m^2\phi^*\phi\right] \\
&= \int \tilde{\mathrm{d}}\boldsymbol{k}\,\tilde{\mathrm{d}}\boldsymbol{q}\,\mathrm{d}^3\boldsymbol{x}\, x^i \Big[k_\mu q^\mu \Big(e^{i(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{x}}\hat{a}_-(k)\hat{a}_+(q) - e^{i(\boldsymbol{k}-\boldsymbol{q})\cdot\boldsymbol{x}}\hat{a}_-(k)\hat{a}^\dagger_-(q) \\
&\quad - e^{-i(\boldsymbol{k}-\boldsymbol{q})\cdot\boldsymbol{x}}\hat{a}^\dagger_+(k)\hat{a}_+(q) + e^{-i(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{x}}\hat{a}^\dagger_+(p)\hat{a}^\dagger_-(q)\Big) - m^2\Big(e^{i(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{x}}\hat{a}_-(k)\hat{a}_+(q) \\
&\quad + e^{i(\boldsymbol{k}-\boldsymbol{q})\cdot\boldsymbol{x}}\hat{a}_-(k)\hat{a}^\dagger_-(q) + e^{-i(\boldsymbol{k}-\boldsymbol{q})\cdot\boldsymbol{x}}\hat{a}^\dagger_+(k)\hat{a}_+(q) + e^{-i(\boldsymbol{k}+\boldsymbol{q})\cdot\boldsymbol{x}}\hat{a}^\dagger_+(p)\hat{a}^\dagger_-(q)\Big)\Big]
\end{aligned}$$

Now we can use the fact that

$$\int \mathrm{d}^3\boldsymbol{x}\, x^i e^{i\boldsymbol{p}\cdot\boldsymbol{x}} = -i(2\pi)^3\frac{\partial}{\partial p_i}\delta^{(3)}(\boldsymbol{p})$$

and also

$$-i(2\pi)^3\int \tilde{\mathrm{d}}\boldsymbol{p}\, f(\boldsymbol{p})\frac{\partial}{\partial p_i}\delta^{(3)}(\boldsymbol{p}) = i(2\pi)^3\int \tilde{\mathrm{d}}\boldsymbol{p}\,\frac{\partial f(\boldsymbol{p})}{\partial p_i}\delta^{(3)}(\boldsymbol{p}) = \frac{i}{2E(\boldsymbol{p})}\left.\frac{\partial f(\boldsymbol{p})}{\partial p_i}\right|_{\boldsymbol{p}=\boldsymbol{0}}$$

Therefore we get

$$\begin{aligned}Q^{[0i]} = \frac{i}{2}\int \frac{\tilde{\mathrm{d}}\boldsymbol{k}}{E(\boldsymbol{k})}\,\Big[&+\tilde{k}_\mu \hat{a}_-(\tilde{k})\frac{\partial}{\partial k_i}\left(k^\mu \hat{a}_+(k)\right) + k_\mu \hat{a}_-(k)\frac{\partial}{\partial k_i}\left(k^\mu \hat{a}_-^\dagger(k)\right)\\ &- k_\mu\,\hat{a}_+^\dagger(k)\frac{\partial}{\partial k_i}\left(k^\mu \hat{a}_+(k)\right) - \tilde{k}_\mu \hat{a}_+^\dagger(\tilde{k})\frac{\partial}{\partial k_i}\left(k^\mu \hat{a}_-^\dagger(k)\right)\Big]\\ &- \frac{i}{2}m^2\int \frac{\tilde{\mathrm{d}}\boldsymbol{k}}{E(\boldsymbol{k})}\,\Big[+\hat{a}_-(\tilde{k})\frac{\partial}{\partial k_i}\hat{a}_+(k) - \hat{a}_-(k)\frac{\partial}{\partial k_i}\hat{a}_-^\dagger(k)\\ &+ \hat{a}_+^\dagger(k)\frac{\partial}{\partial k_i}\hat{a}_+(k) - \hat{a}_+^\dagger(\tilde{k})\frac{\partial}{\partial k_i}\hat{a}_-^\dagger(k)\Big]\end{aligned}$$

Now we have to apply the Leibniz rule

$$\begin{aligned}\frac{\partial}{\partial k_i}\left(k^j \hat{a}(k)\right) &= \frac{\partial k^j}{\partial k_i}\hat{a}(k) + k^j\partial^i\hat{a}(k) = \delta^{ji}\hat{a}(k) + k^j\partial^i\hat{a}(k),\\ \frac{\partial}{\partial k_i}\left(k^0 \hat{a}(k)\right) &= \frac{\partial E(\boldsymbol{k})}{\partial k_i}\hat{a}(k) + E(\boldsymbol{k})\partial^i\hat{a}(k) = \frac{k^i}{E(\boldsymbol{k})}\hat{a}(k) + E(\boldsymbol{k})\partial^i\hat{a}(k),\end{aligned}$$

where these terms are then multiplied by $\frac{1}{E(\boldsymbol{k})}k_0 = \frac{1}{E(\boldsymbol{k})}E(\boldsymbol{k}) = 1$ for the time component and $\frac{1}{E(\boldsymbol{k})}k_j = -\frac{1}{E(\boldsymbol{k})}k^j$ for the space components. We can also rewrite

$$\frac{m^2}{E(\boldsymbol{k})} = E(\boldsymbol{k}) - \frac{\boldsymbol{k}^2}{E(\boldsymbol{k})}.$$

Using this relation after applying the Leibniz rule, we can rewrite our integral in terms of three sums with prefactors $E(\boldsymbol{k})$, $\frac{\boldsymbol{k}^2}{E(\boldsymbol{k})}$ with derivatives of creation / annihilation operators and $\frac{k^i}{E(\boldsymbol{k})}$ without derivatives. Each of these sums will contain eight elements, with each operator pair appearing twice. The operator pairs with one operator depending on $\tilde{k}$ appear always with a different sign and cancel each other out, while the operator pairs depending only on k have different sign only in the $\frac{\boldsymbol{k}^2}{E(\boldsymbol{k})}$ sum, where they cancel, however they have the same sign in the other two sums, where they add up. This way we obtain the final relation:

$$\begin{aligned}Q^{[0i]} = i\int \tilde{\mathrm{d}}k\,\Big[&E(\boldsymbol{k})\left(\hat{a}_-(k)\partial^i\hat{a}_-^\dagger(k) - \hat{a}_+^\dagger(k)\partial^i\hat{a}_+(k)\right)\\ &+ \frac{k^i}{E(\boldsymbol{k})}\left(\hat{a}_-(k)\hat{a}_-^\dagger(k) - \hat{a}_+^\dagger(k)\hat{a}_+(k)\right)\Big]\end{aligned}$$

0.2 Target space symmetries

$\boxed{1.}$ We can see that because the matter matrix of fields is diagonal, any rotation of the fields does not change the Lagrangian density, i.e.:

$$\phi \to R\phi, \qquad \text{where} \quad RR^T = R^T R = \mathbb{1}, R \in \mathbb{R}^{N\times N}$$

therefore the group of the global symmetry is O(N). Here, I am not certain if the continuous global symmetry we are searching means we should only look at its subgroup SO(N), as O(N) also contains elements corresponding to discontinuous symmetries such as reflections?

$\boxed{2.}$ For the generators of this group the following relation must hold

$$M + M^{\mathrm{T}} = 0,$$

i.e. we have the same number of generators as is the number od $N \times N$ antisymmetric real matrices:

$$\dim SO(N) = \frac{1}{2}N(N-1).$$

The infinitesimal transformations of field can be expressed using the Kronecker delta. Rotations can be preformed in planes (ij), where $i, j \in \{1, \dots, N\}, i \neq j$. This fact corresponds to the dimension of the group, as there is $\binom{N}{2} = \frac{1}{2}N(N-1)$ ways to pick two distinct indices (ij) without repetition.

$$\delta_{(ij)}\phi^a = \delta^a_{(i)}\phi^{(j)} - \delta^a_{(j)}\phi^{(i)} = \left[\phi^{(j)}\frac{\partial}{\partial\phi^{(i)}} - \phi^{(i)}\frac{\partial}{\partial\phi^{(j)}}\right]\phi^a$$

For the conserving currents we then have

$$\begin{aligned} J^\mu_{(ij)} &= \sum_{a=1}^{N} \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^a)}\delta_{(ij)}\phi^a + 0 \\ &= \sum_{a=1}^{N} \partial^\mu\phi^a \left[\phi^{(j)}\frac{\partial}{\partial\phi^{(i)}} - \phi^{(i)}\frac{\partial}{\partial\phi^{(j)}}\right]\phi^a \\ &= \partial^\mu\phi^{(i)}\phi^{(j)} - \partial^\mu\phi^{(j)}\phi^{(i)} \end{aligned}$$

$\boxed{3.}$ For the conserved charges we can write

$$Q_{(ij)} = -\int_{t=\text{const.}} \mathrm{d}^3x\, J^0_{(ij)} = -\int_{t=\text{const.}} \mathrm{d}^3x\, \left[\pi^{(i)}\phi^{(j)} - \pi^{(j)}\phi^{(i)}\right]$$

We can now express the field using annihilation and creation operators

$$\begin{aligned}
\phi^a(x) &= \int \tilde{\mathrm{d}}k \left[e^{-ikx}\hat{a}_a(k) + e^{ikx}\hat{a}^\dagger_a(k)\right] \\
\pi^a(x) &= -i\int \tilde{\mathrm{d}}q E(\boldsymbol{q}) \left[e^{-iqx}\hat{a}_a(q) - e^{iqx}\hat{a}^\dagger_a(q)\right]
\end{aligned}$$

By substituting these relations back into the formula for conserved charges and taking $t = 0$, we obtain

$$\begin{aligned}
Q_{(ij)} = & \, i\int \tilde{\mathrm{d}}q\, \tilde{\mathrm{d}}k \mathrm{d}^3x\, E(\boldsymbol{q}) \Big[\Big(e^{i\boldsymbol{q}\cdot\boldsymbol{x}}\hat{a}_{(i)}(q) - e^{-i\boldsymbol{q}\cdot\boldsymbol{x}}\hat{a}^\dagger_{(i)}(q)\Big)\Big(e^{i\boldsymbol{k}\cdot\boldsymbol{x}}\hat{a}_{(j)}(k) + e^{-i\boldsymbol{k}\cdot\boldsymbol{x}}\hat{a}^\dagger_{(j)}(k)\Big) \\
& - \Big(e^{i\boldsymbol{q}\cdot\boldsymbol{x}}\hat{a}_{(j)}(q) - e^{-i\boldsymbol{q}\cdot\boldsymbol{x}}\hat{a}^\dagger_{(j)}(q)\Big)\Big(e^{i\boldsymbol{k}\cdot\boldsymbol{x}}\hat{a}_{(i)}(k) + e^{-i\boldsymbol{k}\cdot\boldsymbol{x}}\hat{a}^\dagger_{(i)}(k)\Big)\Big] \\
= & \, \frac{i}{2}\int \tilde{\mathrm{d}}k \Big[\hat{a}_{(i)}(\tilde{k})\hat{a}_{(j)}(k) + \hat{a}_{(i)}(k)\hat{a}^\dagger_{(j)}(k) - \hat{a}^\dagger_{(i)}(k)\hat{a}_{(j)}(k) - \hat{a}^\dagger_{(i)}(\tilde{k})\hat{a}^\dagger_{(j)}(k) \\
& - \hat{a}_{(j)}(\tilde{k})\hat{a}_{(i)}(k) - \hat{a}_{(j)}(k)\hat{a}^\dagger_{(i)}(k) + \hat{a}^\dagger_{(j)}(k)\hat{a}_{(i)}(k) + \hat{a}^\dagger_{(j)}(\tilde{k})\hat{a}^\dagger_{(i)}(k)\Big] \\
= & \, i\int \tilde{\mathrm{d}}k\, \Big[\hat{a}^\dagger_{(i)}(k)\hat{a}_{(j)}(k) - \hat{a}^\dagger_{(j)}(k)\hat{a}_{(i)}(k)\Big]
\end{aligned}$$